Question ID: f1bfbed3

Question

Marta Coll and colleagues' 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri's 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers' decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____

Which choice most logically completes the text?

Answer

A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's census.

B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.

C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.

D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

Question ID: 6b8a7c74

Question

One recognized social norm of gift giving is that the time spent obtaining a gift will be viewed as a reflection of the gift's thoughtfulness. Marketing experts Farnoush Reshadi, Julian Givi, and Gopal Das addressed this view in their studies of norms specifically surrounding the giving of gift cards, noting that while recipients tend to view digital gift cards (which can be purchased online from anywhere and often can be redeemed online as well) as superior to physical gift cards (which sometimes must be purchased in person and may only be redeemable in person) in terms of usage, 94.8 percent of participants surveyed indicated that it is more socially acceptable to give a physical gift card to a recipient. This finding suggests that _____

Which choice most logically completes the text?

Answer

A. gift givers likely overestimate the amount of effort required to use digital gift cards and thus mistakenly assume gift recipients will view them as less desirable than physical gift cards.

B. physical gift cards are likely preferred by gift recipients because the tangible nature of those cards offers a greater psychological sense of ownership than digital gift cards do.

C. physical gift cards are likely less desirable to gift recipients than digital gift cards are because of the perception that physical gift cards require unnecessary effort to obtain.

D. gift givers likely perceive digital gift cards as requiring relatively low effort to obtain and thus wrongly assume gift recipients will appreciate them less than they do physical gift cards.

Question ID: ce4448b7

Question

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers' reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____

Which choice most logically completes the text?

Answer

A. find the disruptions more irritating as time went on.

B. rate the listening experience as more negative than those whose listening experience was uninterrupted.

C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.

D. perceive the volume of the construction noise as growing softer over time.

Question ID: 4b3d6062

Question

The *Mammillaria* cactus *M. boolii* occurs naturally only in the state of Sonora in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. boolii*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of *M. boolii* could thus _____

Which choice most logically completes the text?

Answer

A. lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.

B. allow conservationists to better consider how climatic changes affecting Sonora may reduce the number of species competing with *M. boolii*.

C. erroneously shift the focus of conservation efforts away from *M. boolii* itself.

D. fail to consider the ways in which the survival of *M. boolii* may be influenced by changes in the populations of other species that inhabit Sonora.

Question ID: 58e9e497

Question

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____

Which choice most logically completes the text?

Answer

A. those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.

B. the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to Haudenosaunee farming techniques.

C. Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.

D. Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the nineteenth century.

Question ID: e185a21f

Question

One theory behind human bipedalism speculates that it originated in a mostly ground-based ancestor that practiced four-legged “knuckle-walking,” like chimpanzees and gorillas do today, and eventually evolved into moving upright on two legs. But recently, researchers observed orangutans, another relative of humans, standing on two legs on tree branches and using their arms for balance while they reached for fruits. These observations may suggest that _____

Which choice most logically completes the text?

Answer

A. bipedalism evolved because it was advantageous to a tree-dwelling ancestor of humans.

B. bipedalism must have evolved simultaneously with knuckle-walking and tree-climbing.

C. moving between the ground and the trees would have been difficult without bipedalism.

D. a knuckle-walking human ancestor could have easily moved bipedally in trees.

Question ID: f9bd4e61

Question

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht's influence can be seen in English playwright Caryl Churchill's 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____

Which choice most logically completes the text?

Answer

A. focus on the characters' beliefs about social and political issues as revealed by the characters' actions.

B. reflect on social and political phenomena not directly related to patriarchy and colonialism.

C. recognize pertinent social and political parallels between Germany during Brecht's time and England at the time when Churchill was writing *Cloud 9*.

D. be dispassionate as they think critically about the social and political questions raised by the play.

Question ID: 4ba0695d

Question

The alpaca was domesticated by Indigenous peoples in the Andes about 7,000 years ago. But which wild species did it descend from, the vicuña or guanaco? A research team led by Ruiwen Fan may have solved the mystery, concluding that the alpaca is the domesticated form of the vicuña but that the modern alpaca gets only 64 percent of its genetic material from its wild ancestor. The rest comes from the domesticated llama. The llama, meanwhile, gets 95.5 percent of its genetic material from its own wild ancestor, the guanaco, and the rest from the alpaca. The llama and alpaca apparently interbred widely for only a handful of generations between 400 and 600 years ago. Assuming that the findings of Fan's team are valid, it can be inferred that _____

Which choice most logically completes the text?

Answer

A. modern llama populations have a greater degree of genetic diversity, on average, than modern alpaca populations do.

B. the domestication process of the alpaca may have involved some introduction of genetic material from the llama.

C. the period of interbreeding resulted in a greater genetic difference between alpacas and their wild ancestors than between llamas and their wild ancestors.

D. if they were subjected to genetic testing, modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding as well.

Question ID: aaddd60f

Question

Scientists studying Mars long thought the history of its crust was relatively simple. One reason for this is that geologic and climate data collected by a spacecraft showed that the crust was largely composed of basalt, likely as a result of intense volcanic activity that brought about a magma ocean, which then cooled to form the planet's surface. A study led by Valerie Payré focused on additional information—further analysis of data collected by the spacecraft and infrared wavelengths detected from Mars's surface—that revealed the presence of surprisingly high concentrations of silica in certain regions on Mars. Since a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica, Payré concluded that _____

Which choice most logically completes the text?

Answer

A. the information about silica concentrations collected by the spacecraft is likely more reliable than the silica information gleaned from infrared wavelengths detected from Mars's surface.

B. high silica concentrations on Mars likely formed from a different process than that which formed the crusts of other planets.

C. having a clearer understanding of the composition of Mars's crust and the processes by which it formed will provide more insight into how Earth's crust formed.

D. Mars's crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.

Question ID: 08395130

Question

The Hubble Space Telescope (HST) is projected to maintain operation until at least 2030, but it has already revolutionized high-resolution imaging of solar-system bodies in visible and ultraviolet (UV) light wavelengths, notwithstanding that only about 6% of the bodies imaged by the HST are within the solar system. NASA researcher Cindy L. Young and colleagues assert that a new space telescope dedicated exclusively to solar-system observations would permit an extensive survey of minor solar-system bodies and long-term UV observation to discern how solar-system bodies change over time. Young and colleagues' recommendation therefore implies that the HST _____

Which choice most logically completes the text?

Answer

A. will likely continue to be used primarily to observe objects outside the solar system.

B. will no longer be used to observe solar system objects if the telescope recommended by Young and colleagues is deployed.

C. can be modified to observe the features of solar system objects that are of interest to Young and colleagues.

D. lacks the sensors to observe the wavelengths of light needed to discern how solar system bodies change over time.

Question ID: 3f236877

Question

Ratified by more than 90 countries, the Nagoya Protocol is an international agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge of wild plants and animals are utilized by agricultural corporations. However, the protocol has shortcomings. For example, it allows corporations to insist that their agreements with communities to conduct research on the commercial uses of the communities' resources and knowledge remain confidential. Therefore, some Indigenous advocates express concern that the protocol may have the unintended effect of _____

Which choice most logically completes the text?

Answer

A. diminishing the monetary reward that corporations might derive from their agreements with Indigenous communities.

B. limiting the research that corporations conduct on the resources of the Indigenous communities with which they have signed agreements.

C. preventing independent observers from determining whether the agreements guarantee equitable compensation for Indigenous communities.

D. discouraging Indigenous communities from learning new methods for harvesting plants and animals from their corporate partners.

Question ID: 95dbdf51

Question

Laura Mulvey has theorized that in narrative film, shots issuing from a protagonist's point of view compel viewers to identify with the character. Such identification is heightened by "invisible editing," or editing so inconspicuous that it renders cuts between shots almost unnoticeable. Conversely, Mulvey proposes that conspicuous editing or an absence of point-of-view shots would induce a more critical stance toward a protagonist. Consider, for example, the attic scene in Alfred Hitchcock's *The Birds*, a conspicuously edited sequence of tens of shots, few of which correspond to the protagonist's point of view. According to Mulvey's logic, this scene should affect viewers by _____

Which choice most logically completes the text?

Answer

A. obscuring their awareness of the high degree of artifice involved in constructing the montage.

B. lessening their identification with the protagonist, if not alienating them from the character altogether.

C. compelling them to identify with the film's director, whose proxy is the camera, and not with the protagonist.

D. diverting their attention away from the film's content and toward its stylistic attributes.

Question ID: 0dba14e6

Question

The increased integration of digital technologies throughout the process of book creation in the late 20th and early 21st centuries lowered the costs of book production, but those decreased costs have been most significant in the manufacturing and distribution process, which occurs after the authoring, editing, and design of the book are complete. This suggests that in the late 20th and early 21st centuries, _____

Which choice most logically completes the text?

Answer

A. digital technologies made it easier than it had been previously for authors to write very long works and get them published.

B. customers generally expected the cost of books to decline relative to the cost of other consumer goods.

C. publishers increased the variety of their offerings by printing more unique titles but also printed fewer copies of each title.

D. the costs of writing, editing, and designing a book were less affected by the technologies used than were the costs of manufacturing and distributing a book.

Question ID: a13c1c66

Question

Many animals, including humans, must sleep, and sleep is known to have a role in everything from healing injuries to encoding information in long-term memory. But some scientists claim that, from an evolutionary standpoint, deep sleep for hours at a time leaves an animal so vulnerable that the known benefits of sleeping seem insufficient to explain why it became so widespread in the animal kingdom. These scientists therefore imply that _____

Which choice most logically completes the text?

Answer

A. prolonged deep sleep is likely advantageous in ways that have yet to be discovered.

B. most traits perform functions that are hard to understand from an evolutionary standpoint.

C. it is more important to understand how widespread prolonged deep sleep is than to understand its function.

D. many traits that provide significant benefits for an animal also likely pose risks to that animal.

Question ID: f942646f

Question

Researchers Suchithra Rajendran and Maximilian Popfinger modeled varying levels of passenger redistribution from short-haul flights (flights of 50 to 210 minutes, from takeoff to landing) to high-speed rail trips. Planes travel faster than trains, but air travel typically requires 3 hours of lead time for security, baggage handling, and boarding that rail travel doesn't, so short-haul routes take similar amounts of time by air and by rail. However, the model suggests that as rail passenger volumes approach current capacity limits, long lead times emerge. Therefore, for rail to remain a viable alternative to short-haul flights, _____

Which choice most logically completes the text?

Answer

A. rail systems should offer fewer long-haul routes and airlines should offer more long-haul routes.

B. rail systems may need to schedule additional trains for these routes.

C. security, baggage handling, and boarding procedures used by airlines may need to be implemented for rail systems.

D. passengers who travel by rail for these routes will need to accept that lead times will be similar to those for air travel.

Question ID: 0dccbf17

Question

Henry Ossawa Tanner's 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore _____

Which choice most logically completes the text?

Answer

A. risk judging Tanner's painting by standards that may not be historically appropriate.

B. tend to conflate Tanner's political views with those of his contemporaries.

C. forgo analyzing Tanner's painting in favor of analyzing his political activity.

D. wrongly assume that Tanner's painting was intended as a critique of his fellow artists.

Question ID: 61228830

Question

A heliograph is a semaphore device used for sending optical communications—usually in the form of Morse code—by reflecting flashes of sunlight off a mirror. Heliographs were used for rapid communication across expansive distances for military, surveying, and forestry purposes during the late nineteenth and early twentieth centuries, but they were largely effective only during the daytime, and the range of the device depended on factors such as the opacity of the air and line of sight. Therefore, heliographs were eventually replaced by technology that _____

Which choice most logically completes the text?

Answer

A. worked on similar principles but was easier to produce and maintain.

B. was not so constrained by environmental circumstances.

C. could be used for more than military, surveying, or forestry purposes.

D. enabled communication that didn’t require knowledge of Morse code.

Question ID: f27559d4

Question

Volunteering, or giving time for a community service for free, is a valuable form of civic engagement because helping in a community is also good for society as a whole. In a survey of youths in the United States, most young people said that they believe volunteering is a way to help people on an individual level. Meanwhile, only 6% of the youths said that they think volunteering is a way to help fix problems in society overall. These replies suggest that _____

Which choice most logically completes the text?

Answer

A. many young people think they can volunteer only within their own communities.

B. volunteering may be even more helpful than many young people think it is.

C. volunteering can help society overall more than it can help individual people.

D. many young people may not know how to find ways to volunteer their time.

Question ID: d1539546

Question

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh's interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____

Which choice most logically completes the text?

Answer

A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.

B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs' location.

C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.

D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

Question ID: 9abc3ba5

Question

“Gestures” in painting are typically thought of as bold, expressive brushstrokes. In the 1970s, American painter Jack Whitten built a 12-foot (3.7-meter) tool he named the “developer” to apply paint to an entire canvas in one motion, resulting in his series of “slab” paintings from that decade. Whitten described this process as making an entire painting in “one gesture,” signaling a clear departure from the prevalence of gestures in his work from the 1960s. Some art historians claim this shift represents “removing gesture” from the process. Therefore, regardless of whether using the developer constitutes a gesture, both Whitten and these art historians likely agree that _____

Which choice most logically completes the text?

Answer

A. any tool that a painter uses to create an artwork is capable of creating gestures.

B. Whitten’s work from the 1960s exhibits many more gestures than his work from the 1970s does.

C. Whitten became less interested in exploring the role of gesture in his work as his career progressed.

D. Whitten’s work from the 1960s is much more realistic than his work from the 1970s is.

Question ID: 5c7e0d62

Question

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Vita Sancti Cadoci*, composed in the 11th century, and *Culhwch and Olwen* from the second half of the 12th century. One of the most significant sources, Geoffrey of Monmouth's *History of the Kings of Britain*, was written in the 1130s; some material from it was later adapted by the Norman poet Wace into the *Roman de Brut* in 1155. But Wace didn't merely adapt *History*, he added to it as well, introducing the famous Round Table at which Arthur's knights assembled, which suggests that _____

Which choice most logically completes the text?

Answer

A. *Roman de Brut* is more historically accurate than *History,* because *Culhwch and Olwen* had not been written when Geoffrey of Monmouth was writing his work.

B. Geoffrey of Monmouth was unaware of stories of the Round Table when composing his *History*, though historians know that works containing such stories were available to him.

C. Geoffrey of Monmouth's accounts of Arthurian legends in his *History* are more similar overall in content to the accounts in *Culhwch and Olwen* than they are to the accounts in *Roman de Brut*.

D. the Arthurian legends that the author of *Vita Sancti Cadoci* drew on would not have featured the Round Table.

Question ID: 22b3da87

Question

During the Bourbon Restoration in France (1814–1830), the right to vote required in part that a person paid at least 300 francs in direct taxes to the government. The four most common taxes (the *quatre vieilles*) were levied on real estate (both land and buildings); the doors and windows in taxpayer homes; the rental values of homes; and the businesses of artisans and merchants. (Foreign investments were either exempt from taxation or taxed lightly.) Although relatively few people paid the tax on real estate, it was the main means of voter qualification and accounted for over two-thirds of government receipts during this period, suggesting that during the Bourbon Restoration _____

Which choice most logically completes the text?

Answer

A. those people who had the right to vote most likely had substantial holdings of French real estate.

B. the voting habits of French artisans and merchants were effective in reducing tax burdens on businesses.

C. the number of doors and windows in French residences was kept to a minimum but increased after 1830.

D. French people with significant foreign investments were unlikely to have the right to vote.

Question ID: e503ae04

Question

With the ongoing expansion of e-commerce, consumers are expecting faster and faster delivery of goods, but delivery companies continue to struggle with last-mile logistics (the final step in delivery to consumers) due to challenges such as complex and inefficient delivery routes. Innovations to mitigate these challenges have been emerging—the use of aerial drones, for example—but these innovations tend to engender their own complications (e.g., regulations on the use of drones in residential airspace), leading researchers to conclude that _____

Which choice most logically completes the text?

Answer

A. consumers' expectations for reduced delivery times may be outstripping what is viable for delivery companies to provide.

B. a better understanding of consumers' expectations for delivery is needed so that companies can better plan for fluctuations in delivery volume.

C. rapid delivery is a leading factor in consumer satisfaction, and therefore delivery companies would benefit from investing resources in reducing delivery times.

D. there may not be sufficient incentive for delivery companies to attempt to solve the problems associated with last-mile logistics.

Question ID: cae97f58

Question

Mosses can struggle in harsh desert conditions because these plants require enough sunlight for photosynthesis but not so much that they risk drying out. Researchers Jenna Ekwealor and Kirsten M. Fisher found several species of *Syntrichia caninervis*, a type of desert moss, growing under quartz crystals in California's Mojave Desert. To evaluate whether these semitransparent rocks benefited the moss, the researchers compared the shoot tissue, a measure of plant growth, of *S. caninervis* when growing on the soil surface versus when the moss was growing under the quartz rocks. They found that the shoot tissue was 62% longer for moss growing under the quartz as compared to moss on the soil surface, suggesting that _____

Which choice most logically completes the text?

Answer

A. *S. caninervis* is one of the few types of moss that can survive under semitransparent rocks.

B. quartz crystals do not transmit the necessary sunlight for photosynthesis in *S. caninervis*.

C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.

D. quartz crystals are capable of supporting *S. caninervis* growth if the crystals are not too thin.

Question ID: 65502c46

Question

In a 2017 article, historian Jasmine Kilburn-Toppin explains that in early modern London, members of the city's guilds (trade and artisanal associations) were participants in a civic culture in which gift giving both signaled and conferred social status. Research on this phenomenon has tended to focus on philanthropic gifting by London's largest guilds; for her part, Kilburn-Toppin focuses on the gifting of handmade objects and fixtures (such as decorative paneling or plasterwork) within the craft guilds, which were "composed of highly discerning producers and consumers of material cultures." Given this characterization, it can reasonably be inferred that the gifting of such objects may have _____

Which choice most logically completes the text?

Answer

A. ensured that knowledge of the materials and techniques used by members of the craft guilds stayed within the craft guilds.

B. emphasized ostentatious offerings in the case of the largest guilds and prosaic objects in the case of the craft guilds.

C. functioned as a way for craft guild members to maintain and enhance their professional reputations among their peers.

D. conferred greater social status when the recipient belonged to one of London's largest guilds than when the recipient belonged to a craft guild.

Question ID: 03701ef3

Question

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps' burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow's measurements both at a moment in time and throughout the entire burrow-building process, something that's impossible with resin casting because _____

Which choice most logically completes the text?

Answer

A. it can only be used on burrows below a certain size.

B. it does not allow for multiple castings of the same burrow over time.

C. the casting process takes more time than *A. bellulus* takes to construct a burrow.

D. the process of recovering the model distorts the resin's shape.